Nonhomogeneous Equations & Undetermined Coefficients

ANSWER KEY

Section 1: General Solution Structure

1 What is the structure of the general solution to a linear nonhomogeneous equation $L[y] = g(x)$?

a. $y(x) = y_c(x) + y_p(x)$**, where** y_c **is the complementary solution and** y_p **is any particular solution** ✓

b. $y(x) = y_c(x) \cdot y_p(x)$

c. $y(x) = y_p(x)$ only

d. $y(x) = c_1 y_p(x) + c_2 y_c(x)$

By linearity, the general solution of a nonhomogeneous linear equation is the sum of the complementary solution y_c to the homogeneous equation and any particular solution y_p.

2 Find a particular solution y_p to $y'' - 5y' + 6y = 12$.

a. $y_p = 2$ ✓

b. $y_p = 12$

c. $y_p = 2x$

d. $y_p = 6$

Try constant $y_p = A$. Then $y_p' = 0$, $y_p'' = 0$. Substituting: $6A = 12 \Rightarrow A = 2$.

3 What is the trial particular solution y_p for $y'' + 4y = 3x^2 - 1$?

a. $y_p = Ax^2 + Bx + C$ ✓

b. $y_p = Ax^2$

c. $y_p = Ax^2 + B$

d. $y_p = Ax^3 + Bx^2$

For a quadratic polynomial forcing term $g(x) = 3x^2 - 1$, the standard trial solution is a complete quadratic polynomial $y_p = Ax^2 + Bx + C$.

Section 2: Exponential and Sinusoidal Forcing Functions

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4 Find a particular solution to $y'' + y = 4e^{2x}$.

a. $y_p = \frac{4}{5}e^{2x}$ ✓

b. $y_p = 4e^{2x}$

c. $y_p = 2e^{2x}$

d. $y_p = \frac{1}{2}e^{2x}$

Try $y_p = Ae^{2x}$. $y_p'' = 4Ae^{2x}$. Substitute: $4Ae^{2x} + Ae^{2x} = 5Ae^{2x} = 4e^{2x} \Rightarrow 5A = 4 \Rightarrow A = 4/5$.

5 What is the correct undetermined coefficients form for $y'' + 9y = 5\cos(2x)$?

a. $y_p = A\cos(2x) + B\sin(2x)$ ✓

b. $y_p = A\cos(2x)$

c. $y_p = Ax\cos(2x)$

d. $y_p = A\sin(2x)$

Because $\cos(2x)$ produces both sines and cosines upon differentiation, the complete form is $A\cos(2x) + B\sin(2x)$.

6 Find y_p for $y'' + 9y = 5\cos(2x)$.

a. $y_p = \cos(2x)$ ✓

b. $y_p = 5\cos(2x)$

c. $y_p = \frac{5}{13}\cos(2x)$

d. $y_p = \sin(2x)$

Let $y_p = A\cos(2x)$. $y_p'' = -4A\cos(2x)$.
$-4A\cos(2x) + 9A\cos(2x) = 5A\cos(2x) = 5\cos(2x) \Rightarrow 5A = 5 \Rightarrow A = 1$.

Section 3: The Modification Rule (Resonance / Overlap)

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7 When must a trial particular solution y_p be multiplied by x (or x^2)?

a. **When any term in the trial solution is already a solution of the associated homogeneous equation ✓**

b. Whenever the forcing function is a polynomial

c. When the coefficient of y' is negative

d. Only when the right-hand side is zero

If a term in the proposed y_p is part of the complementary solution y_c, substituting it will yield zero; multiplying by x restores linear independence.

8 For $y'' - 4y' + 4y = 3e^{2x}$, the homogeneous solution is $y_c = (c_1 + c_2x)e^{2x}$. What is the correct trial form for y_p?

a. $y_p = Ax^2e^{2x}$ ✓

b. $y_p = Ae^{2x}$

c. $y_p = Axe^{2x}$

d. $y_p = Ax^3e^{2x}$

Both e^{2x} and xe^{2x} are in y_c. Multiplying by x^2 produces $y_p = Ax^2e^{2x}$, which is linearly independent of y_c.

9 What physical phenomenon occurs when an undamped mechanical system is driven at its natural resonance frequency ($y'' + \omega_0^2 y = F_0\cos(\omega_0 t)$)?

a. **Resonance, where amplitude grows linearly with time ($y_p = Ct\sin(\omega_0 t)$) ✓**

b. Immediate decay to zero

c. Critical damping

d. Chaotic motion

Because the forcing frequency equals the natural frequency, the particular solution takes the form $t\sin(\omega_0 t)$, producing unbounded linearly growing oscillations (resonance).

Section 4: The Principle of Superposition

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10 If y_{p1} solves $L[y] = g_1(x)$ and y_{p2} solves $L[y] = g_2(x)$, what solves $L[y] = 2g_1(x) - 3g_2(x)$?

a. $2y_{p1} - 3y_{p2}$ ✓

b. $y_{p1}y_{p2}$

c. $\frac{y_{p1}}{y_{p2}}$

d. $y_{p1} + y_{p2}$

By the linearity of differential operators, $L[2y_{p1} - 3y_{p2}] = 2L[y_{p1}] - 3L[y_{p2}] = 2g_1(x) - 3g_2(x)$.

11 Can the Method of Undetermined Coefficients be used to solve $y'' + y = \tan x$?

a. **No, because derivatives of $\tan x$ produce infinitely many non-terminating forms; Variation of Parameters is required** ✓

b. Yes, with trial solution $A\tan x + B\sec x$

c. Yes, with trial solution $A\sin x/\cos x$

d. Only if x is small

Undetermined coefficients applies only to functions with finite dimensional spaces of derivatives (polynomials, exponentials, sines/cosines). For $\tan x$, variation of parameters must be used.

12 Find a particular solution to $y'' + 4y = 8x + 12$.

a. $y_p = 2x + 3$ ✓

b. $y_p = 2x$

c. $y_p = 4x + 3$

d. $y_p = 2x^2 + 3x$

Let $y_p = Ax + B$. $y_p'' = 0$. $4(Ax + B) = 4Ax + 4B = 8x + 12 \Rightarrow 4A = 8 \Rightarrow A = 2$ and $4B = 12 \Rightarrow B = 3$. Thus $y_p = 2x + 3$.

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